

PLSC 504 – Fall 2026

Time Series Analysis: An Introduction

September 28, 2026

Longitudinal Data = Data reflecting temporal variation.

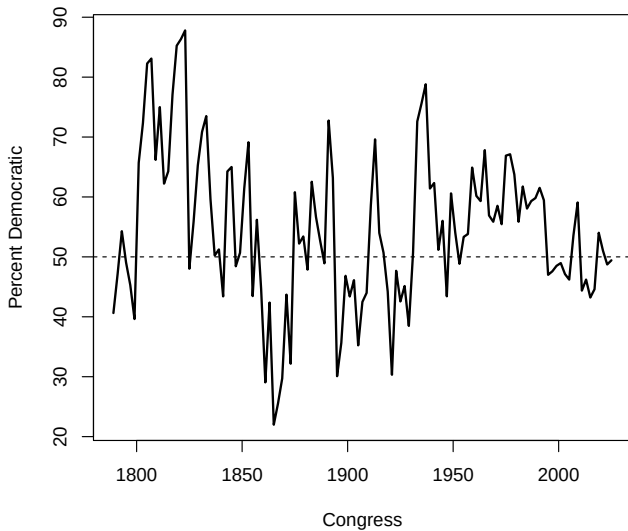
Typically:

- $i \in \{1, 2, \dots, N\}$ indexes cross-sectional units
- $t \in \{1, 2, \dots, T\}$ indexes temporal observations

Terminology:

- Cross-Sectional: N is large, $T = 1$
- Time Series: T is large, $N = 1$
- Panel: N is large, T is small(ish)
- TSCS: $N \approx T$, or $N < T$ (but $N > 1$)

Democratic House Membership, 1789-2025



- **Univariate:** Properties of series, ARIMA, etc.
- **Bivariate:** Pairs of series / regression
 - “Distributed Lag” models
 - Structural breaks / intervention analysis
 - Cointegration / Error Correction
 - Others...
- **Multivariate:** Systems of equations
 - “Seemingly unrelated” regressions
 - Vector Autoregressions (“VARs”) and their variants
- **Higher-moment properties:** ARCH, GARCH, etc.

Consider a random series $\mathbf{Y}$:

$$\mathbf{Y} = \{Y_0, Y_1, \dots, Y_T\}$$

with moments:

$$\begin{aligned}\mu_t &= E(Y_t) \\ \sigma_t^2 \equiv \text{Var}(Y_t) &= E[(Y_t - \mu_t)^2] \\ \gamma_{t,t-s} \equiv \text{Cov}(Y_t, Y_{t-s}) &= E[(Y_t - \mu_t)(Y_{t-s} - \mu_{t-s})]\end{aligned}$$

Two typical assumptions: stationarity and ergodicity.

Stationarity: A constant d.g.p. over time.¹

Mean stationarity:

$$E(Y_t) = \mu \forall t$$

Variance stationarity:

$$\text{Var}(Y_t) = E[(Y_t - \mu)^2] \equiv \sigma_Y^2 \forall t$$

Covariance stationarity:

$$\text{Cov}(Y_t, Y_{t-s}) = E[(Y_t - \mu)(Y_{t-s} - \mu)] = \gamma_s \forall s$$

¹A stricter form of stationarity requires that the joint probability distribution (in other words, *all* the moments) of series of observations $\{Y_1, Y_2, \dots, Y_t\}$ is the same as that for $\{Y_{1+s}, Y_{2+s}, \dots, Y_{t+s}\}$ for all t and s .

Ergodicity: Asymptotic (in T) independence.

Means that:

$$\text{Cov}(Y_t, Y_{t-s}) = \gamma_s \rightarrow 0 \text{ as } s \rightarrow \infty$$

Stationary ergodic processes:

- are measure-preserving,
- can have their statistical properties inferred/estimated from a sufficiently long observation of the process.

Autocovariance and Autocorrelation

For a stationary ergodic series:

$$\hat{\mu} = \bar{Y} = T^{-1} \sum_{t=1}^T Y_t$$

$$\hat{\sigma}^2 = T^{-1} \sum_{t=1}^T (Y_t - \bar{Y})^2$$

$$\hat{\gamma}_s = T^{-1} \sum_{t=s+1}^T (Y_t - \bar{Y})(Y_{t-s} - \bar{Y}), \quad s = 1, 2, 3, \dots$$

Autocovariance $\rightarrow$ Autocorrelation:

$$\hat{\rho}_s = \frac{\hat{\gamma}_s}{\hat{\sigma}^2}, \quad s = 0, \pm 1, \pm 2, \dots$$

The “ARIMA” Approach

“ARIMA” = Autoregressive Integrated Moving Average...

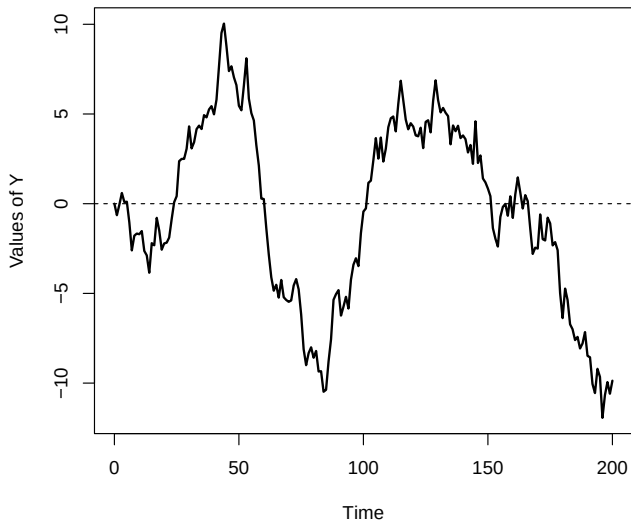
A (first-order) integrated series (“random walk”) is:

$$Y_t = Y_{t-1} + u_t, \quad u_t \sim i.i.d.(0, \sigma_u^2)$$

...a/k/a a “random walk”:

$$\begin{aligned} Y_t &= Y_{t-2} + u_{t-1} + u_t \\ &= Y_{t-3} + u_{t-2} + u_{t-1} + u_t \\ &= \sum_{t=0}^T u_t \end{aligned}$$

I(1) series with $u_t \sim N(0, 1)$, $T = 200$



I(1) series are not stationary.

Variance:

$$\text{Var}(Y_t) \equiv E(Y_t)^2 = t\sigma^2$$

Autocovariance:

$$\text{Cov}(Y_t, Y_{t-s}) = |t-s|\sigma^2.$$

Both depend on t ...

I(1) Series: Differencing

For an I(1) series:

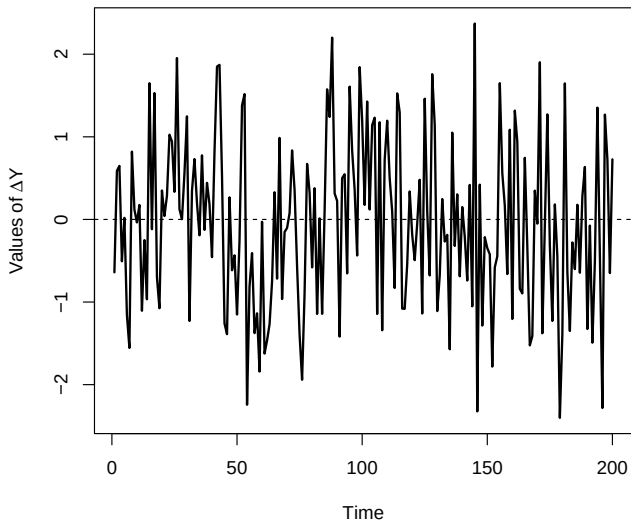
$$Y_t - Y_{t-1} = u_t$$

which we often write in terms of the difference operator Δ (or sometimes ∇):

$$\Delta Y_t = u_t$$

The differenced series is just the (stationary, ergodic) white-noise process u_t .

A Differenced I(1) Series



Higher-order $I(d)$ Integrated Series

An $I(d)$ series is one which, differenced d times, becomes a stationary series.

E.g., the $I(2)$ series:

$$Y_t = u_t + 2u_{t-1} + 3u_{t-2} + \dots$$

Differencing:

$$\begin{aligned}\Delta Y_t &= [u_t + 2u_{t-1} + 3u_{t-2} + \dots] - [u_{t-1} + 2u_{t-2} + 3u_{t-3} + \dots] \\ &= \sum_{j=0}^T u_{t-j}\end{aligned}$$

Differencing again:

$$\begin{aligned}\Delta^2 Y_t \equiv (Y_t - Y_{t-1}) - (Y_{t-1} - Y_{t-2}) &= [u_t + u_{t-1} + \dots] - [u_{t-1} + u_{t-2} + \dots] \\ &= u_t\end{aligned}$$

which is stationary / ergodic.

An AR(1) process:

$$Y_t = \phi Y_{t-1} + u_t$$

with $u_t \sim i.i.d.(0, \sigma_u^2)$.

Equivalently:

$$Y_t - \phi Y_{t-1} = u_t$$

or:

$$\begin{aligned} Y_t &= \phi[\phi Y_{t-2} + u_{t-1}] + u_t \\ &= \phi^2 Y_{t-2} + \phi u_{t-1} + u_t \\ &= \phi^2[\phi Y_{t-3} + u_{t-2}] + \phi u_{t-1} + u_t \\ &= \phi^3 Y_{t-3} + \phi^2 u_{t-2} + \phi u_{t-1} + u_t \\ &= \dots \\ &= \sum_{j=0}^{T-1} \phi^j u_{t-j} + \phi^T Y_0 \end{aligned}$$

Mean:

$$\begin{aligned} E(Y_t) &= E\left(\sum_{j=0}^{T-1} \phi^j u_{t-j}\right) + E(\phi^T Y_{t-T}) \\ &= \phi^T Y_0 \end{aligned}$$

That means that:

- For $|\phi| > 1$, $|E(Y_t)|$ is increasing in t
- For $|\phi| = 1$, $E(Y_t) = Y_0$
- For $|\phi| < 1$:
 - the importance of Y_0 decreases over time
 - asymptotically, $E(Y_t) = 0$

AR(1) and Stationarity

For an AR(1) series with $|\phi| < 1$:

$$Y_t = \sum_{j=0}^{\infty} \phi^j u_{t-j}$$

Means that:

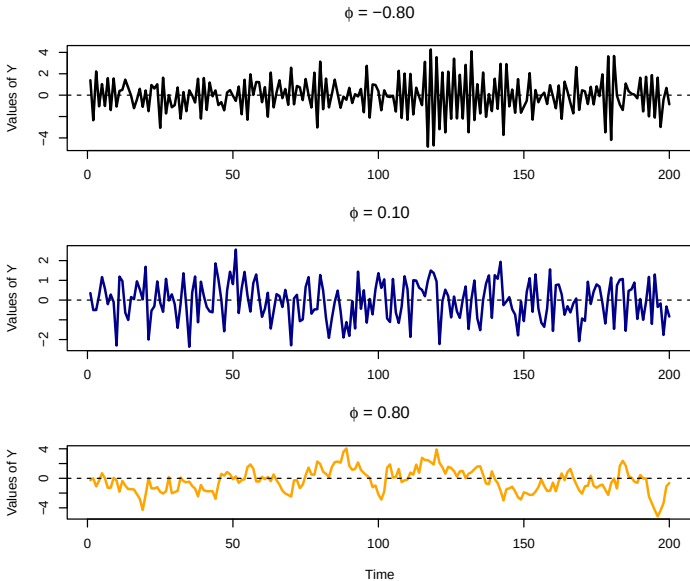
- the effects of a shock *persist* over time; however,
- the *size* of those effects *decay* over time
- In the limit, the effect of any one shock very much later is zero

Similarly:

$$\begin{aligned}\text{Var}(Y_t) \equiv E(Y_t^2) &= E\left(\sum_{j=0}^{\infty} \phi^j u_{t-j}\right)^2 \\ &= \sum_{j=0}^{\infty} \phi^{2j} E(u_{t-j}^2) \\ &= \sigma^2 \sum_{j=0}^{\infty} \phi^{2j} \\ &= \frac{\sigma^2}{1 - \phi^2}\end{aligned}$$

which is stationary iff $|\phi| < 1$.

AR(1) Series, Illustrated



Notation Digression: The Lag Operator

“Lag operator” L :

$$LY_t = Y_{t-1}$$

$$L^2 Y_t = Y_{t-2}$$

$$L^s Y_t = Y_{t-s}$$

$$L^0 Y_t = Y_t$$

Means we can write:

$$\begin{aligned}\Delta Y_t &= Y_t - Y_{t-1} \\ &= (1 - L)Y_t\end{aligned}$$

So an AR(1) equation is:

$$(1 - \phi L)Y_t = u_t$$

Higher-Order AR(p) Series

An AR(p) series:

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + u_t$$

Equivalently:

$$\begin{aligned} Y_t - \phi_1 Y_{t-1} - \phi_2 Y_{t-2} - \dots - \phi_p Y_{t-p} &= u_t \\ (1 - \phi L - \phi_2 L^2 - \dots - \phi_p L^p) Y_t &= u_t \end{aligned}$$

Moving Average (MA) Series

MA(1) series:

$$Y_t = u_t + \theta u_{t-1}, \quad t = 1, 2, \dots, T$$

with $u_t \sim \text{i.i.d.}(0, \sigma_u^2)$.

Mean:

$$\begin{aligned}\mu &= E(Y_t) \\ &= E(u_t) + \theta E(u_{t-1}) \\ &= 0\end{aligned}$$

Variance:

$$\begin{aligned}\sigma_Y^2 &= E[(u_t + \theta u_{t-1})(u_t + \theta u_{t-1})] \\ &= E(u_t^2) + \theta^2 E(u_{t-1}^2) + 2\theta E(u_t u_{t-1}) \\ &= (1 + \theta^2) \sigma_u^2\end{aligned}$$

Autocovariance at one lag:

$$\begin{aligned}\gamma_1 &= E[(u_t + \theta u_{t-1})(u_{t-1} + \theta u_{t-2})] \\ &= E(u_t u_{t-1}) + \theta E(u_{t-1}^2) + \theta E(u_t u_{t-2}) + \theta^2 E(u_{t-1} u_{t-2}) \\ &= \theta E(u_{t-1}^2) \\ &= \theta \sigma_u^2\end{aligned}$$

Autocovariance at $s > 1$ lags:

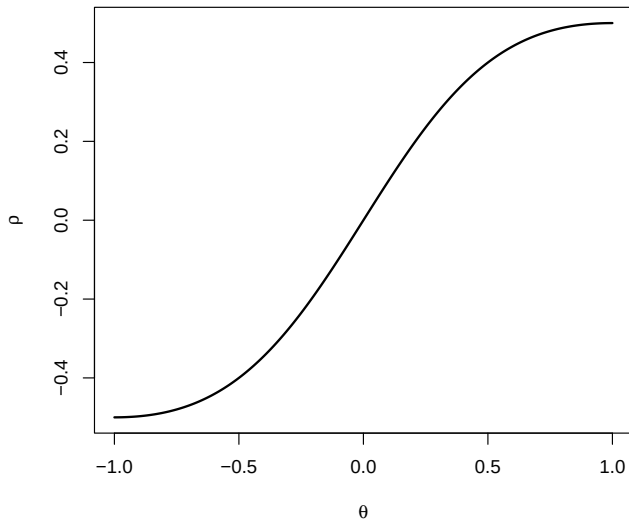
$$\begin{aligned}\gamma_s &= E[(u_t + \theta u_{t-s})(u_{t-1} + \theta u_{t-(s+1)})] \\ &= E(u_t u_{t-1}) + \theta E(u_{t-s}^2) + \theta E(u_t u_{t-(s+1)}) + \theta^2 E(u_{t-s} u_{t-(s+1)}) \\ &= 0\end{aligned}$$

This means that:

- means + variances + autocovariances are all independent of t ,
- the ACF is zero for all lags greater than one,
- the ACF at one lag is dependent on the degree of dependence in the moving average process. In particular, since $\rho = \frac{\gamma}{\sigma^2}$:

$$\begin{aligned}\rho &= \frac{\theta\sigma_u^2}{(1 + \theta^2)\sigma_u^2} \\ &= \frac{\theta}{1 + \theta^2}\end{aligned}$$

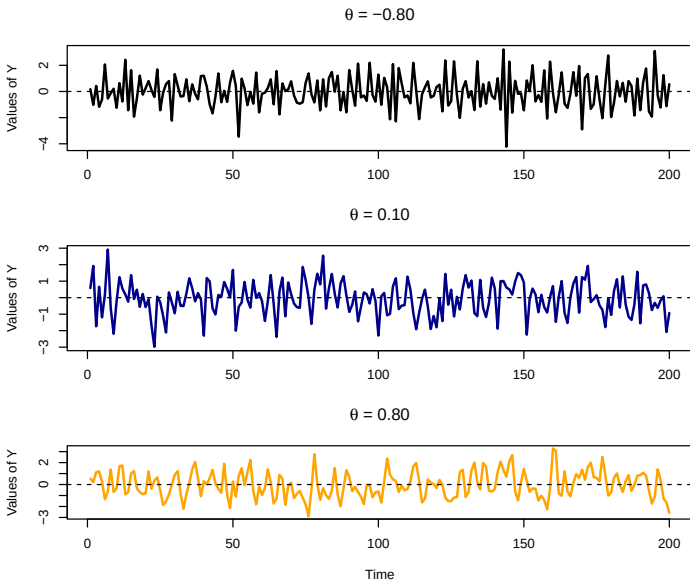
Relationship between θ and ρ for an MA(1) series



For an MA(1) series:

- $\theta = 0 \leftrightarrow \rho_1 = 0$, a “white noise” process (no temporal dependence in Y_t).
- For $\theta > 0$:
 - $\rho_1 > 0$
 - successive values of Y_t will be *positively* related, and
 - the series will be “smoother” than a white noise series.
- For $\theta < 0$:
 - $\rho_1 < 0$
 - successive values of Y_t will be *negatively* correlated, and
 - the series will be less “smooth” than a white noise sequence.

MA(1) Series Illustrated



Rewriting our MA(1) series:

$$\begin{aligned}
 Y_t &= u_t + \theta u_{t-1} \\
 &= u_t + \theta(Y_{t-1} - \theta u_{t-2}) \\
 &= u_t + \theta Y_{t-1} - \theta^2 u_{t-2} \\
 &= u_t + \theta Y_{t-1} - \theta^2(Y_{t-2} - \theta u_{t-3}) \\
 &= u_t + \theta Y_{t-1} - \theta^2 Y_{t-2} + \theta^3 u_{t-3} \\
 &= \dots \\
 &= u_t + \theta Y_{t-1} - \theta^2 Y_{t-2} + \theta^3 Y_{t-3} - \dots - (-\theta)^{(T-1)} Y_{t-(T-1)} - (-\theta)^T u_0
 \end{aligned}$$

As $T \rightarrow \infty$:

$$Y_t = - \sum_{j=1}^{\infty} (-\theta)^j Y_{t-j} + u_t$$

The Wold Decomposition Theorem

Any weakly stationary, purely nondeterministic stochastic process can be written as a linear combination of a sequence of uncorrelated random variables.

that is, as:

$$Y_t = \sum_{j=0}^{\infty} \psi_j u_{t-j}$$

with $\sum_{j=0}^{\infty} \psi_j^2 < \infty$.

MA(q) model:

$$Y_t = u_t + \theta_1 u_{t-1} + \theta_2 u_{t-2} + \dots + \theta_q u_{t-q}$$

MA(q) models are:

- always stationary,
- invertable if the AR(p) form of the model satisfies the stationarity conditions above.

ARMA / ARIMA Models

A general ARMA(p, q) model is:

$$Y_t = \phi_1 Y_{t-1} + \dots + \phi_p Y_{t-p} + u_t + \theta_1 u_{t-1} + \dots + \theta_q u_{t-q}$$

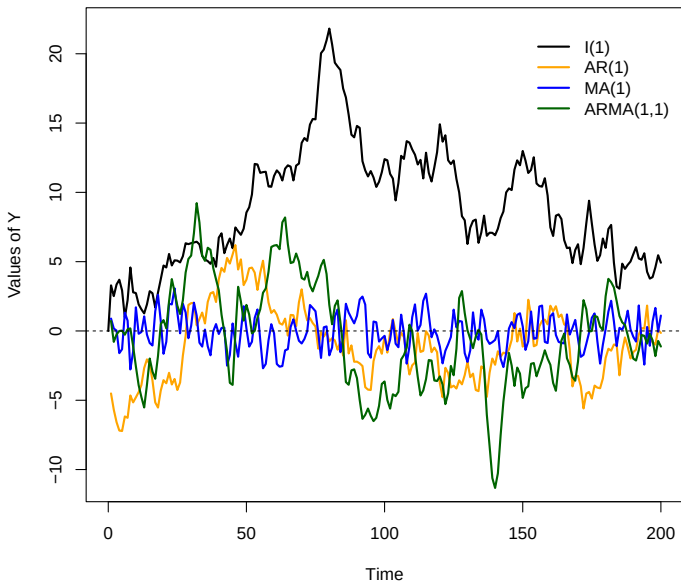
Equivalently:

$$(1 - \phi L - \phi_2 L^2 - \dots - \phi_p L^p) Y_t = (1 + \theta L + \theta_2 L^2 + \dots + \theta_q L^q) u_t$$

If the model also requires d th-order differencing in order to achieve stationarity, we call it an ARIMA(p, d, q) model; e.g., the ARIMA($p, 1, q$) model:

$$\Delta Y_t = \phi_1 Y_{t-1} + \dots + \phi_p Y_{t-p} + u_t + \theta_1 u_{t-1} + \dots + \theta_q u_{t-q}$$

ARIMA Series: A Comparison ($\phi = \theta = 0.9$)



Series Characteristics: ACFs and PACFs

Autocorrelation function (ACF):

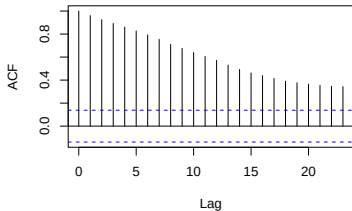
$$\hat{\rho}_s = \frac{\hat{\gamma}_s}{\hat{\sigma}^2}, \quad s = 0, \pm 1, \pm 2, \dots$$

The partial autocorrelation function (PACF) is the correlation between Y_t and Y_{t-s} after controlling for (“partialling out”) the the common linear effects of the intermediate lags.²

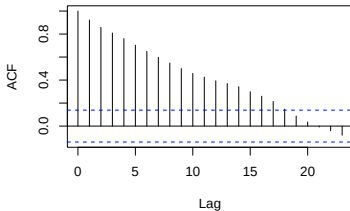
²The PACF is estimated from a solution to the Yule-Walker system of equations. A good mathematical treatment of this is in Box et al. 1994, pp. 64-69

ARIMA Series: ACFs

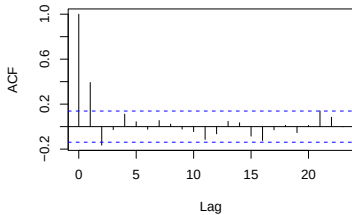
I(1) series



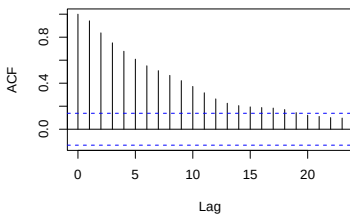
AR(1) series



MA(1) series

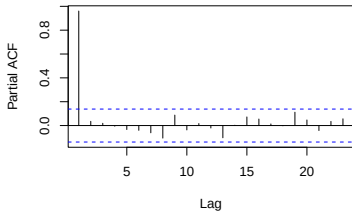


ARMA(1,1) series

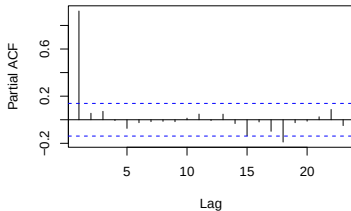


ARIMA Series: PACFs

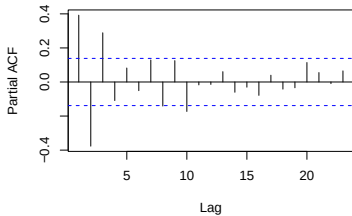
I(1) series



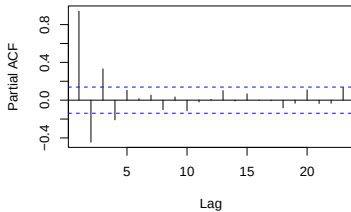
AR(1) series



MA(1) series



ARMA(1,1) series



ARIMA Modeling: Intuition

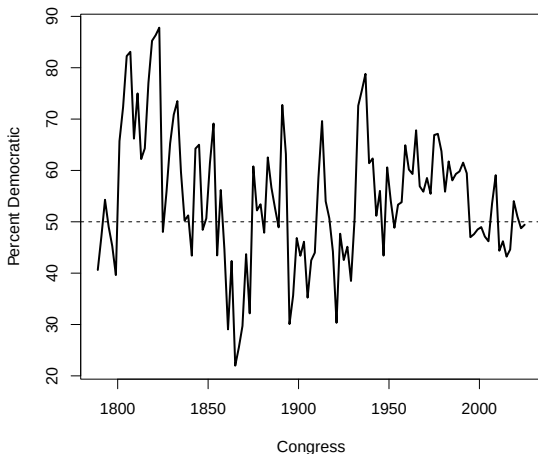
Identification $\rightarrow$ *Estimation* $\rightarrow$ *Diagnostics*

In general:

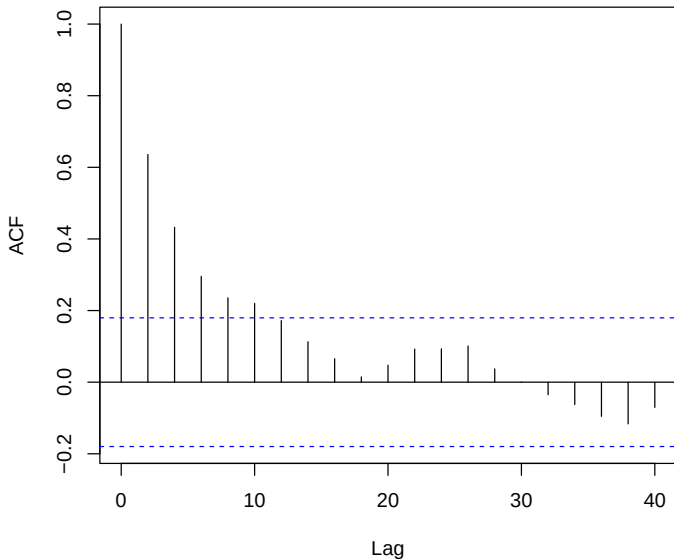
1. Determine if the series is stationary. If so, proceed to (3).
2. If not, difference the series until it is.
3. Examine the ACFs and PACFs of the stationary series, in order to determine the data generating process (AR, MA, or some combination thereof).
4. Fit a model – starting simple – using ARIMA/MLE.
5. Examine and test the residuals to determine if they are white noise. If they are, proceed to (7).
6. If they are not, go back to (3) and try again.
7. Proceed with inference, hypothesis testing, forecasting, and the like.

Example: Democratic House Membership

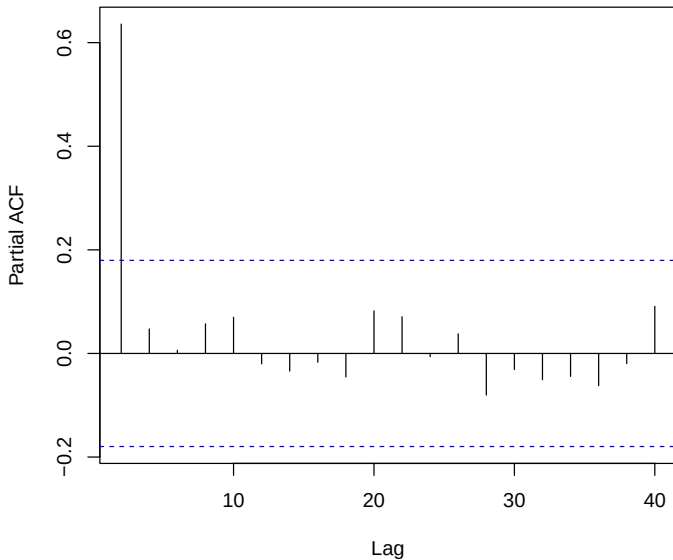
Y = percentage of U.S. House members identifying with the Democratic party, by Congress, 1789-2025.



Democratic House Membership: ACF



Democratic House Membership: PACF



ARIMA Model Fitting

```
> DH.AR1 <- arima(DH.TS,order=c(1,0,0),method="ML") # AR(1)
> summary(DH.AR1)
```

Call:

```
arima(x = DH.TS, order = c(1, 0, 0), method = "ML")
```

Coefficients:

	ar1	intercept
	0.638	54.35
s.e.	0.070	2.48

sigma² estimated as 98.8: log likelihood = -442, aic = 891

Training set error measures:

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	0.1	9.94	7.81	-3.76	15.7	0.143	-0.0315

```
> DH.ARMA11 <- arima(DH.TS,order=c(1,0,1),method="ML") # ARMA(1,1)
> summary(DH.ARMA11)
```

Call:

```
arima(x = DH.TS, order = c(1, 0, 1), method = "ML")
```

Coefficients:

	ar1	ma1	intercept
	0.682	-0.076	54.3
s.e.	0.105	0.146	2.6

sigma² estimated as 98.5: log likelihood = -442, aic = 892

Training set error measures:

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	0.109	9.93	7.77	-3.73	15.6	0.142	-0.000927

ARIMA Model Selection

```
> # Model selection via LR test:
>
> lrtest(DH.AR1,DH.ARMA11)
Likelihood ratio test

Model 1: arima(x = DH.TS, order = c(1, 0, 0), method = "ML")
Model 2: arima(x = DH.TS, order = c(1, 0, 1), method = "ML")
      #Df LogLik Df Chisq Pr(>Chisq)
1      3   -442
2      4   -442  1  0.27      0.6

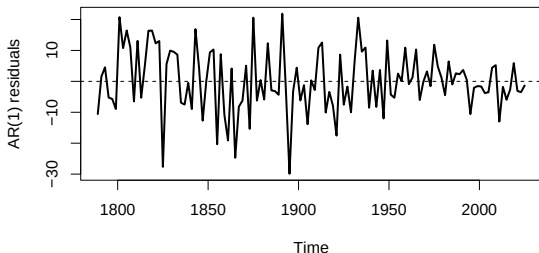
> # Automated version:
>
> DH.robot <- auto.arima(DH.TS)
> summary(DH.robot)
Series: DH.TS
ARIMA(1,0,0) with non-zero mean

Coefficients:
          ar1      mean
          0.638  54.35
s.e.      0.070   2.48

sigma^2 = 100: log likelihood = -442
AIC=891   AICc=891   BIC=899

Training set error measures:
              ME RMSE MAE  MPE MAPE  MASE   ACF1
Training set 0.1 9.94 7.81 -3.76 15.7 0.143 -0.0315
```

ARIMA: Residuals Analysis



Box-Pierce and Ljung-Box tests:

```
> Box.test(DH.AR1$residuals)
```

Box-Pierce test

```
data: DH.AR1$residuals
```

```
X-squared = 0.1, df = 1, p-value = 0.7
```

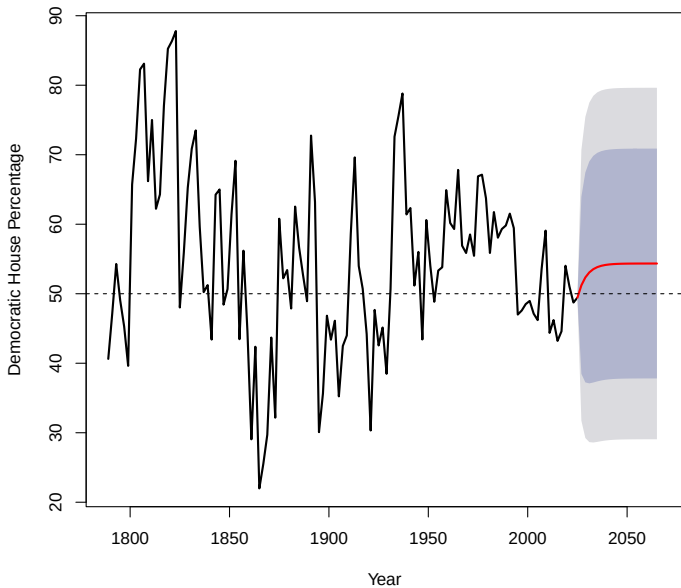
```
> Box.test(DH.AR1$residuals,type="Ljung")
```

Box-Ljung test

```
data: DH.AR1$residuals
```

```
X-squared = 0.1, df = 1, p-value = 0.7
```

ARIMA: Forecasting (40-Year Window)



Unit Roots, (Co)Integration, and Error Correction

A basic $I(1)$ series is:

$$Y_t = Y_{t-1} + u_t$$

Special case of AR(1):

$$Y_t = \rho Y_{t-1} + u_t$$

where $\rho = 1$.

Recall:

- $|\rho| < 1.0 \leftrightarrow$ stationary series
- $|\rho| = 1.0 \leftrightarrow$ non-stationary series

Other Non-Stationary Series

Trending series:

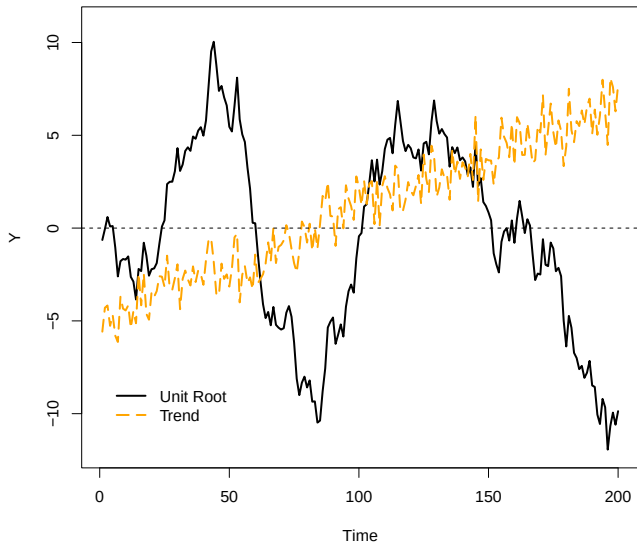
$$Y_t = \beta t + u_t$$

Means:

$$\begin{aligned}\Delta Y_t \equiv Y_t - Y_{t-1} &= \beta t + u_t - Y_{t-1} \\ &= \beta(t) + u_t - \beta(t-1) - u_{t-1} \\ &= \beta(t - t + 1) + u_t - u_{t-1} \\ &= \beta + \Delta u_t\end{aligned}$$

which is also stationary.

Unit Root vs. Trend (same u_i s)



Key: Examine

$$Y_t = \rho Y_{t-1} + \beta t + u_t$$

and test for $H_0 : \hat{\beta} = 0$:

- Cannot reject $\hat{\beta} = 0 \leftrightarrow$ “random walk” without a trend;
- Reject $\hat{\beta} = 0 \leftrightarrow$ series has a deterministic trend.

For $Y_t = \rho Y_{t-1} + u_t$:

- $|\rho| > 1$
 - Series is nonstationary / *explosive*
 - Past shocks have a greater impact than current ones
 - Uncommon
- $|\rho| < 1$
 - *Stationary* series
 - Effects of shocks die out exponentially according to ρ
 - Is mean-reverting
- $|\rho| = 1$
 - Nonstationary series
 - Shocks persist at full force
 - Not mean-reverting; variance increases with t

Basic idea:

$$Y_t = \rho Y_{t-1} + u_t$$

and test for $\hat{\rho} = 1$.

Issues:

- $\hat{\rho} = \frac{\sum Y_t Y_{t-1}}{\sum Y_{t-1}^2}$ is consistent for ρ , but
- the distribution of $\hat{\rho}$ is not t ...
 - $T(\hat{\rho} - 1) \rightarrow \frac{\int_0^1 W(r) dW(r)}{\int_0^1 W(r)^2 dr}$, so
 - $t \equiv \frac{\hat{\rho}}{s.e.(\hat{\rho})} = \frac{\frac{1}{2}[W(1)^2 - 1]}{[\int_0^1 W(r)^2 dr]^{\frac{1}{2}}}$

This is:

- Right-skewed (so t -statistics will tend to be large and negative),
- $\rightarrow$ tend to *overreject* the null hypothesis if we use the standard t -distribution.

The Dickey-Fuller Test

To test for a unit root:

1. Estimate $\hat{\rho}$,
2. Test $H_0 : \hat{\rho} = 1$ using a non-standard set of critical values,
3. Ensure that $\hat{u}$ are white noise.

Equivalent:

$$\begin{aligned}\Delta Y_t &= (\rho - 1)Y_{t-1} + u_t \\ &= \delta Y_{t-1} + u_t\end{aligned}$$

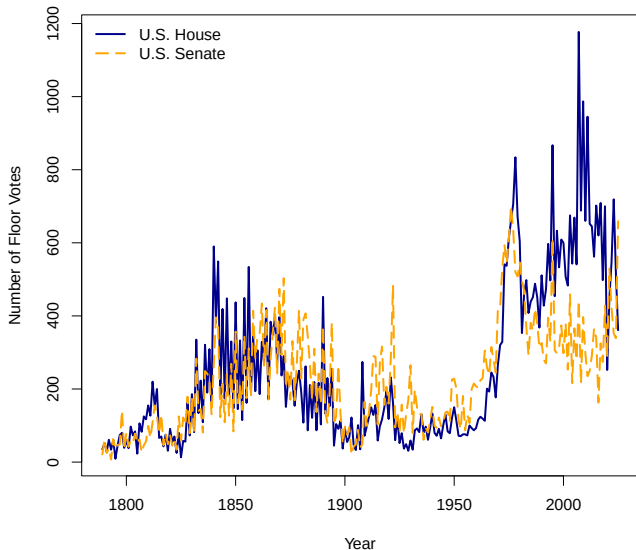
and test for $\hat{\delta} = 0$.

Dickey-Fuller Variants + Other Tests

Unit root testing:

- Dickey-Fuller Variants:
 - Add “drift”: $Y_t = \alpha + \rho Y_{t-1} + u_t$
 - Add a trend: $Y_t = \alpha + \beta t + \rho Y_{t-1} + u_t$
 - “Augmented” D-F tests: $\Delta Y_t = \rho Y_{t-1} + \sum_{i=1}^p d_i \Delta Y_{t-i} + u_t$
- Phillips-Perron test
- Kwiatkowski et al. (“KPSS”):
- “Variance Ratio” tests
- Others...

House and Senate Votes, 1789-2025



Basic Dickey-Fuller Test

```
> HDF<-ur.df(HVotes.TS,type="none",lags=0)
> summary(HDF)
```

```
#####
# Augmented Dickey-Fuller Test Unit Root Test #
#####
```

Call:

```
lm(formula = z.diff ~ z.lag.1 - 1)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-384.0	-41.8	18.3	79.4	685.5

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
z.lag.1	-0.0915	0.0276	-3.31	0.0011 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 142 on 235 degrees of freedom

Multiple R-squared: 0.0446, Adjusted R-squared: 0.0405

F-statistic: 11 on 1 and 235 DF, p-value: 0.00108

Value of test-statistic is: -3.31

Critical values for test statistics:

	1pct	5pct	10pct
tau1	-2.58	-1.95	-1.62

D-F Tests with Drift/Trend

```
> HDF.D<-ur.df(HVotes.TS,type="drift",lags=0)
> summary(HDF.D)
```

```
#####
# Augmented Dickey-Fuller Test Unit Root Test #
#####
```

Value of test-statistic is: -5.355 14.35

Critical values for test statistics:

	1pct	5pct	10pct
tau2	-3.46	-2.88	-2.57
phi1	6.52	4.63	3.81

```
> HDF.T<-ur.df(HVotes.TS,type="trend",lags=0)
> summary(HDF.T)
```

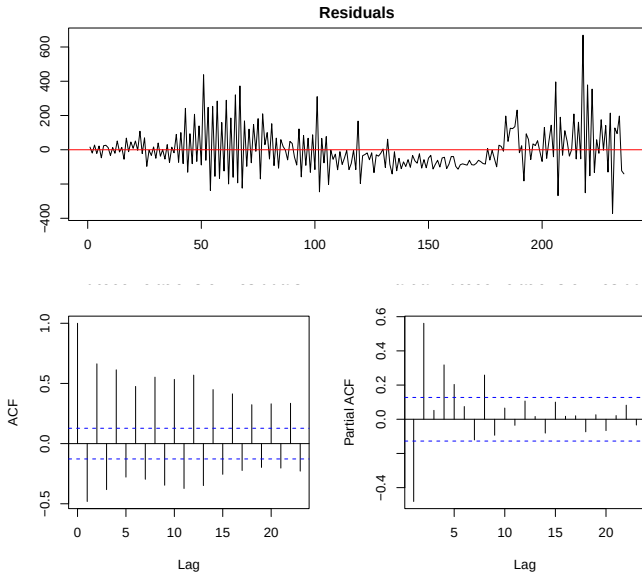
```
#####
# Augmented Dickey-Fuller Test Unit Root Test #
#####
```

Value of test-statistic is: -6.689 14.92 22.37

Critical values for test statistics:

	1pct	5pct	10pct
tau3	-3.99	-3.43	-3.13
phi2	6.22	4.75	4.07
phi3	8.43	6.49	5.47

Plots: Residuals + P/ACFs



“Augmented” D-F Test

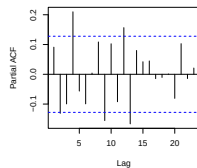
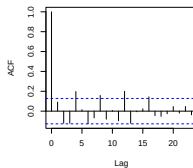
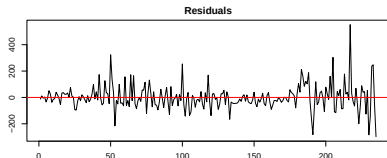
```
> HADF.T1<-ur.df(HVotes.TS,type="trend",lags=1)
> summary(HADF.T1)
```

```
#####
# Augmented Dickey-Fuller Test Unit Root Test #
#####
```

Value of test-statistic is: -2.497 2.153 3.12

Critical values for test statistics:

	1pct	5pct	10pct
tau3	-3.99	-3.43	-3.13
phi2	6.22	4.75	4.07
phi3	8.43	6.49	5.47



ADF Test: Lag Selection via BIC

```
> HADF.BIC<-ur.df(HVotes.TS,type="trend",lags=6,selectlags="BIC")
> summary(HADF.BIC)
```

```
#####
# Augmented Dickey-Fuller Test Unit Root Test #
#####
```

```
Call:
lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-299.2	-44.0	-12.0	33.8	537.2

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6.4682	12.6463	0.51	0.60953
z.lag.1	-0.0784	0.0373	-2.10	0.03683 *
tt	0.1403	0.1145	1.22	0.22195
z.diff.lag1	-0.5965	0.0701	-8.51	2.5e-15 ***
z.diff.lag2	-0.0512	0.0814	-0.63	0.52995
z.diff.lag3	-0.2598	0.0665	-3.90	0.00012 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 91.7 on 224 degrees of freedom

Multiple R-squared: 0.62, Adjusted R-squared: 0.611

F-statistic: 73 on 5 and 224 DF, p-value: <2e-16

Value of test-statistic is: -2.1 1.596 2.207

Critical values for test statistics:

	1pct	5pct	10pct
tau3	-3.99	-3.43	-3.13
phi2	6.22	4.75	4.07
phi3	8.43	6.49	5.47

Packages:

- `tseries`
- `uroot` (including seasonality)
- `urca`
- `egcm`
- `CADFtest`

Consider:

$$Y_t = \rho Y_{t-1} + u_t$$

Properties:

1. Linear transformations of Y_t preserve the level of integration.
2. For two $I(0)$ series X_t and Y_t , $aX_t + bY_t + c$ is also $I(0)$.
3. For two $I(1)$ series X_t and Y_t , $aX_t + bY_t + c$ is *generally* $I(1)$ as well.
4. If X_t is $I(0)$ and Y_t is $I(1)$, then $aX_t + bY_t + c$ is $I(1)$ (*integration dominates stationarity*).

Now consider:

$$\begin{aligned}X_t &= W_t + u_{Xt} \\Y_t &= AW_t + u_{Yt}, \\W_t &\sim I(1), \\u_{Xt}, u_{Yt} &\sim I(0)\end{aligned}$$

Here, both X_t and Y_t will be $I(1)$, but:

$$\begin{aligned}Z_t &= Y_t - AX_t \\&= [AW_t + u_{Yt}] - A[W_t + u_{Xt}] \\&= AW_t + u_{Yt} - AW_t - Au_{Xt} \\&= u_{Yt} - Au_{Xt}\end{aligned}$$

is $I(0)$.

If there exist two $I(1)$ series X_t, Y_t such that

$$Z_t = \mu + aX_t + bY_t \text{ is } I(0),$$

then X_t and Y_t are said to be cointegrated.

Intuition:

- The nonstationarity in X and Y arises from a *common component* (W_t) which is $I(1)$.
- Combining X and Y cancels out the common part, leaving $I(0)$ “residuals.”

Cointegration as an Attractor

Above:

$$X_t = AY_t$$

means that:

$$Z_t = X_t - AY_t \sim \text{i.i.d.}(0, \sigma_Z^2)$$

Examples:

1. Murray (1994): Drunks and puppies...
2. Regional commodity prices...

Cointegration in Practice

First: Assess whether X and Y are $I(1)$.

Then rewrite:

$$Z_t = X_t - \alpha Y_t$$

as the cointegrating regression:

$$X_t = \alpha Y_t + Z_t.$$

Cointegration implies:

$$\begin{aligned} Z_t &\sim I(0), \text{ and} \\ \text{Var}(Z_t) &< \infty. \end{aligned}$$

Cointegration: A Roadmap

1. Determine that the two series are $I(1)$,
2. Estimate $X_t = \hat{\alpha} Y_t + Z_t$ using OLS,
3. Examine $\hat{Z}_t$ for stationarity, using
 - a Durbin-Watson test (“CRDW”), and/or
 - standard unit root tests (DF, ADF, KPSS, etc.)

Wait, What? OLS???

Recall that for cointegrated X_t and Y_t , $\text{Var}(Z_t) < \infty$.

Suppose we (incorrectly) estimate δ :

$$\tilde{Z}_t = X_t - \delta Y_t$$

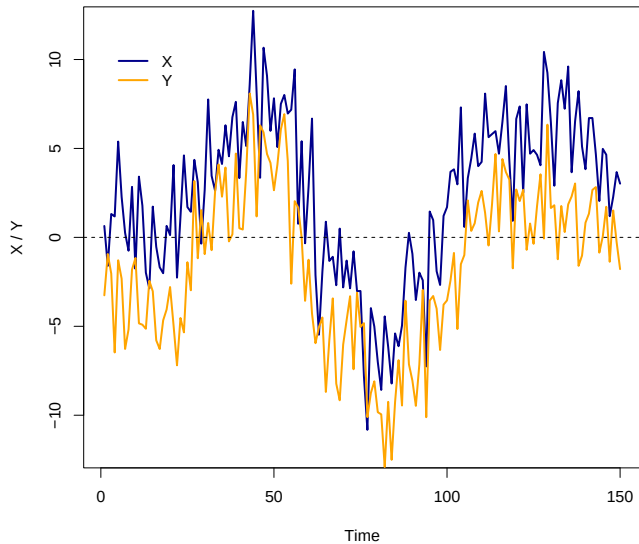
Then $\tilde{Z}_t$ is $I(1)$, and $\lim_{T \rightarrow \infty} [\text{Var}(\tilde{Z}_t)] = \infty$, while in finite samples:

$$\text{Var}(\tilde{Z}_t) > \text{Var}(\hat{Z}_t),$$

Thus:

- OLS minimizes residual variance $\rightarrow$
- OLS makes $\hat{Z}_t$ stationary [and makes $\text{Var}(Z_t)$ finite] $\rightarrow$
- $\hat{\alpha}_{OLS} \rightarrow \alpha$ as $T \rightarrow \infty$ (“superconsistent”).

A Simulated Example



Unit Root Tests

```
> summary(ur.df(X.TS,type="trend",lags=1))
```

```
.  
.   
.
```

```
Value of test-statistic is: -3.262 3.566 5.335
```

```
Critical values for test statistics:
```

	1pct	5pct	10pct
tau3	-3.99	-3.43	-3.13
phi2	6.22	4.75	4.07
phi3	8.43	6.49	5.47

```
> summary(ur.df(Y.TS,type="trend",lags=1))
```

```
.  
.   
.
```

```
Value of test-statistic is: -2.683 2.4 3.6
```

```
Critical values for test statistics:
```

	1pct	5pct	10pct
tau3	-3.99	-3.43	-3.13
phi2	6.22	4.75	4.07
phi3	8.43	6.49	5.47

CI Regression + Residual Check

```
> CI.reg <- lm(X~Y)    # cointegrating regression
> Zhats.TS <- ts(CI.reg$residuals,start=1,end=T)
> summary(ur.df(Zhats.TS,type="trend",lags=1))
```

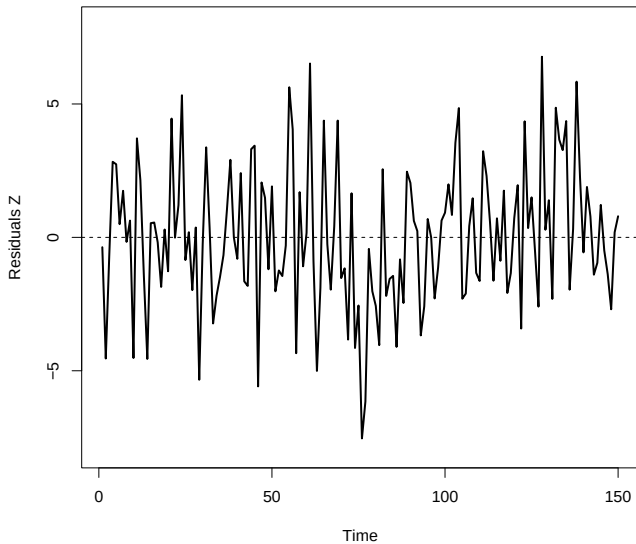
```
.
.
.
```

Value of test-statistic is: -9.06 27.38 41.06

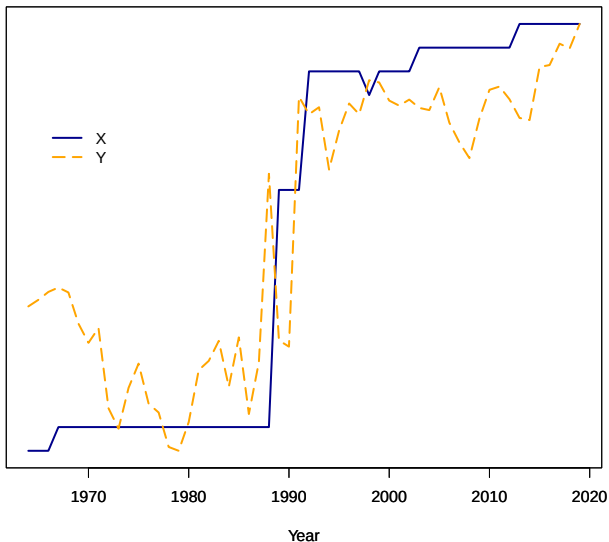
Critical values for test statistics:

	1pct	5pct	10pct
tau3	-3.99	-3.43	-3.13
phi2	6.22	4.75	4.07
phi3	8.43	6.49	5.47

Residuals from Cointegrating Regression



Two (More) Series, 1964-2019



A Regression

```
> summary(lm(SY.TS~SX.TS))
```

Call:

```
lm(formula = SY.TS ~ SX.TS)
```

Residuals:

Min	1Q	Median	3Q	Max
-3751	-806	108	598	4657

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	21178	194	108.9	<2e-16 ***
SX.TS	404	25	16.2	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1450 on 54 degrees of freedom

Multiple R-squared: 0.829, Adjusted R-squared: 0.826

F-statistic: 261 on 1 and 54 DF, p-value: <2e-16

Another Regression (Lagged X)

```
> summary(dyn$lm(SY.TS~lag(SX.TS,-1)))
```

Call:

```
lm(formula = dyn(SY.TS ~ lag(SX.TS, -1)))
```

Residuals:

Min	1Q	Median	3Q	Max
-3864	-788	138	595	4613

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	21277.2	188.4	113.0	<2e-16 ***
lag(SX.TS, -1)	410.5	24.3	16.9	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1400 on 53 degrees of freedom

(2 observations deleted due to missingness)

Multiple R-squared: 0.844, Adjusted R-squared: 0.841

F-statistic: 286 on 1 and 53 DF, p-value: <2e-16



X = Ford Mustang prices, in
constant \$US



Y = Paraguay's POLITY IV
(democracy / autocracy)
score

Lagged X s:

$$Y_t = \alpha + \beta X_{t-1} + u_t$$

often yield autocorrelated errors / spurious regressions.

...vs. “difference equations”:

$$\Delta Y_t = \alpha + \beta \Delta X_{t-1} + u_t$$

where all effects are “short term.”

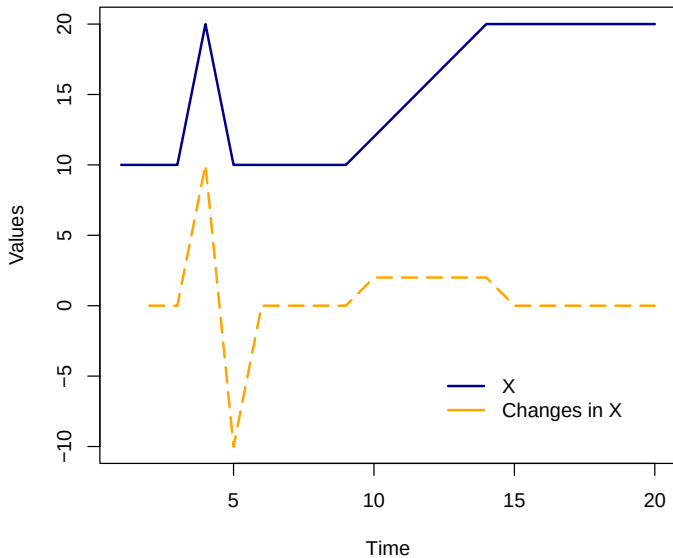
“ECMs”:

$$\Delta Y_t = \beta \Delta X_{t-1} + \rho(Y_{t-1} - \alpha - \gamma X_{t-1})$$

Terms:

- $\hat{\beta} \rightarrow$ short-term relationship between X and Y .
- $\hat{\alpha}$ and $\hat{\gamma} \rightarrow$ long-term relationship between X and Y (the “attractor” – the equilibrium distance between X and Y).
- $\hat{\rho} \rightarrow$ the rate at which the model “reequilibrates” (formally, the proportion of the disequilibrium which is corrected with each period)

ECM: Toy Example



Two Cointegrated Series

Two examples:

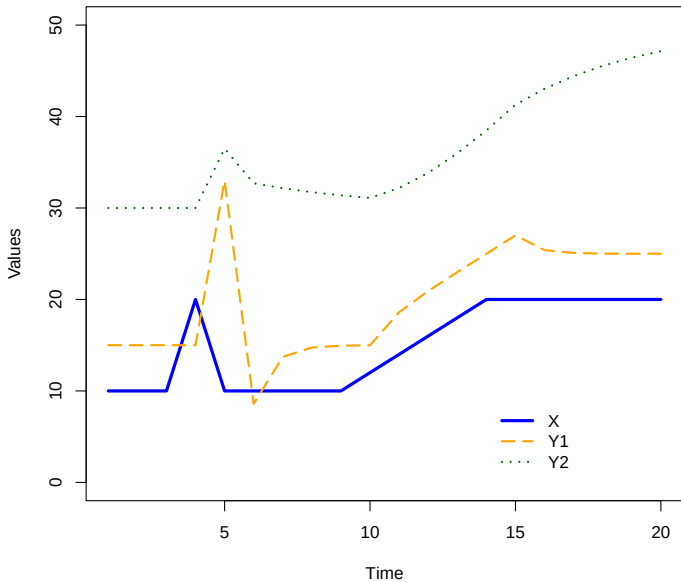
$$\Delta Y_{1t} = 1.0\Delta X_{t-1} - 0.8(Y_{1t-1} - 5.0 - 1.0X_{t-1})$$

$$\Delta Y_{2t} = 0.25\Delta X_{t-1} - 0.2(Y_{2t-1} - 10.0 - 2.0X_{t-1})$$

Note:

- ΔY_1 has a larger degree of short-term dependence on X .
- The “equilibrium distance” for ΔY_1 is $Y_t = 5 + X_t$; that in ΔY_2 is $Y_t = 10 + 2X_t$.
- Y_1 reequilibrates at a much faster rate than does Y_2 .

A Picture!



Estimation and Testing: Two-Steps...

If both X_t and Y_t are cointegrated, then a general ECM is:

$$\begin{aligned}\Delta Y_t &= \beta_0 + \beta_1 \Delta X_{t-1} + \beta_2 \Delta X_{t-2} + \dots + \beta_k \Delta X_{t-k} + \rho Z_{t-1} + u_t \\ &= \beta_0 + \beta_1 \Delta X_{t-1} + \beta_2 \Delta X_{t-2} + \dots + \beta_k \Delta X_{t-k} + \rho(X_t - \alpha Y_t) + u_t\end{aligned}$$

where the ΔX s capture the short-term dependence between X and Y .

The Engle-Granger Two-Step:

1. Estimate the cointegrating regression $Y_t = \alpha + \gamma X_t + e_t$,
2. From these estimates, generate $\hat{Z}_t = Y_t - \hat{\alpha} - \hat{\gamma} X_t$,
3. Include $\hat{Z}_{t-1}$ for Z_{t-1} in the model above.

The One-Step Approach

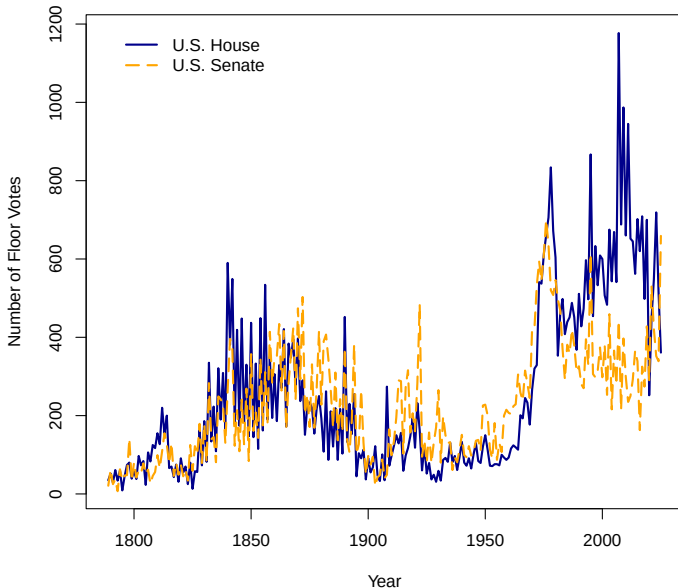
Substituting $Z_t = X_t - \alpha Y_t$ into the equation above, we get:

$$\begin{aligned}\Delta Y_t &= \beta_0 + \beta_1 \Delta X_{t-1} + \rho(Y_{t-1} - \alpha - \gamma X_{t-1}) + u_t \\ &= (\beta_0 - \rho\alpha) + \beta_1 \Delta X_{t-1} + \rho Y_{t-1} - \rho\gamma X_{t-1} + u_t\end{aligned}$$

Suggests a single equation, where changes in Y_t are a function of:

- Lagged ΔX s,
- X_{t-1} , and
- Y_{t-1} .

Example: House and Senate Votes, 1789-2025



Unit Roots? (A: Yes)

```
> summary(ur.df(HVotes.TS,type="trend",lags=6,selectlags="BIC"))
```

Value of test-statistic is: -2.1 1.596 2.207

Critical values for test statistics:

	1pct	5pct	10pct
tau3	-3.99	-3.43	-3.13
phi2	6.22	4.75	4.07
phi3	8.43	6.49	5.47

```
> summary(ur.df(SVotes.TS,type="trend",lags=6,selectlags="BIC"))
```

Value of test-statistic is: -2.388 2.157 2.874

Critical values for test statistics:

	1pct	5pct	10pct
tau3	-3.99	-3.43	-3.13
phi2	6.22	4.75	4.07
phi3	8.43	6.49	5.47

ECM, Step One

```
> StepOne <- lm(SVotes.TS~HVotes.TS) # CI regression  
> summary(StepOne)
```

Call:

```
lm(formula = SVotes.TS ~ HVotes.TS)
```

Residuals:

Min	1Q	Median	3Q	Max
-326.7	-66.5	-16.8	51.9	380.6

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	103.2950	9.9393	10.4	<2e-16 ***
HVotes.TS	0.4851	0.0297	16.3	<2e-16 ***

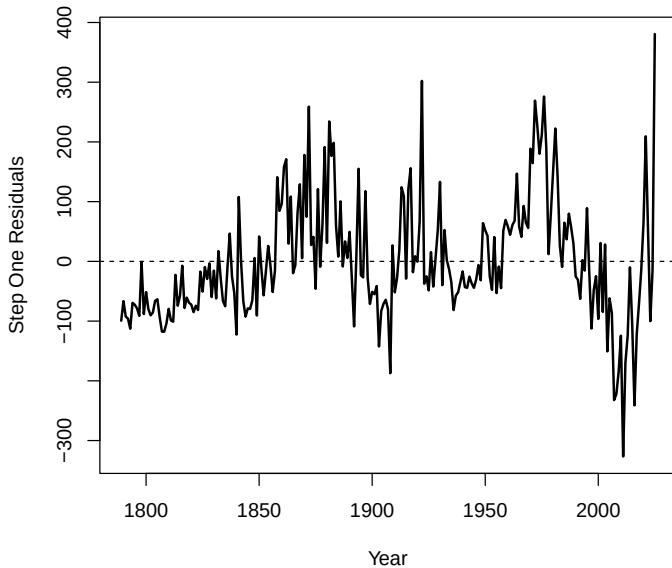
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 101 on 235 degrees of freedom

Multiple R-squared: 0.531, Adjusted R-squared: 0.529

F-statistic: 266 on 1 and 235 DF, p-value: <2e-16

Step One Residuals...☹



ECM: Step Two

```
> DSV.TS <- diff(SVotes.TS)           # difference Y
> DHVLag.TS <- lag(diff(HVotes.TS),k=-1) # Lagged differenced X
> Ztminus1.TS <- lag(Zt.TS,k=-1)       # Lagged residuals
> df <- ts.intersect(DSV.TS,DHVLag.TS,Ztminus1.TS)

> StepTwo <- lm(DSV.TS ~ DHVLag.TS + Ztminus1.TS,
+               data = df)
> summary(StepTwo)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.9278	5.8997	0.50	0.62
DHVLag.TS	-0.4041	0.0407	-9.92	< 2e-16 ***
Ztminus1.TS	-0.3674	0.0605	-6.07	0.000000005 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 90.4 on 232 degrees of freedom

Multiple R-squared: 0.373, Adjusted R-squared: 0.368

F-statistic: 69.1 on 2 and 232 DF, p-value: <2e-16

What Does That Mean?

The fitted ECM is:

$$\Delta \text{Senate Votes}_t = 2.92 - 0.40 \Delta \text{House Votes}_{t-1} - 0.37[\text{Senate Votes}_{t-1} - 103.3 - 0.49(\text{House Votes}_{t-1})]$$

This means that:

- The “short term” relationship between Senate and House votes has about 0.40.
- The “equilibrium distance” between Senate and House floor vote totals is roughly $\text{Senate Votes} = 103 - 0.5 (\text{House Votes})$.
- Following a “shock,” the relationship between the two returns to its equilibrium level at a rate of about 37 percent per year.

One-Step ECM

```
> SVLag.TS <- lag(SVotes.TS,k=-1) # Lag Y
> HVLag.TS <- lag(HVotes.TS,k=-1) # Lag X
> df2 <- ts.intersect(DSV.TS,DHVLag.TS,SVLag.TS,HVLag.TS)
> OneStep <- lm(DSV.TS~DHVLag.TS+SVLag.TS+HVLag.TS, data=df2)

> summary(OneStep)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	46.3653	11.1989	4.14	0.0000486694	***
DHVLag.TS	-0.3933	0.0432	-9.11	< 2e-16	***
SVLag.TS	-0.3684	0.0606	-6.08	0.0000000048	***
HVLag.TS	0.1573	0.0404	3.89	0.00013	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 90.5 on 231 degrees of freedom

Multiple R-squared: 0.375, Adjusted R-squared: 0.367

F-statistic: 46.2 on 3 and 231 DF, p-value: <2e-16

A Potpourri of Other Topics...

- Distributed lag models
- (“Granger”) Causality
- Intervention analysis, “structural breaks,” and threshold models
- Vector autoregression (VAR) models
- Autoregressive Conditional Heteroscedasticity (ARCH and GARCH) models
- Fractional integration / long-memoried series
- Spectral domain analysis
- Time series models for discrete (binary, ordered, event count, etc.) outcomes
- Models for spatio-temporal data

Some favorite references:

- Box-Steffensmeier, Janet M., John R. Freeman, Matthew P. Hitt, and Jon C. Pevehouse. 2015. *Time Series Analysis for the Social Sciences*. New York: Cambridge University Press.
- Fuller, W.A. 1996. *Introduction to Statistical Time Series*, 2nd ed. New York: Wiley.
- Pickup, Mark. 2014. *Introduction to Time Series Analysis*. New York: SAGE Publications. Quantitative Applications in the Social Sciences.
- Shumway, Robert H., and David S. Stoffer. 2025. *Time Series Analysis and Its Applications, With R Examples*, 5th Ed. New York: Springer.

Syllabi (examples; some also include panel data models):

- [Adolph \(2026\)](#)
- [Brandt \(2024\)](#)
- [Enns \(2023\)](#)
- [Mitchell \(2020\)](#)
- [Philips \(2022\)](#)